

Not so SMEs

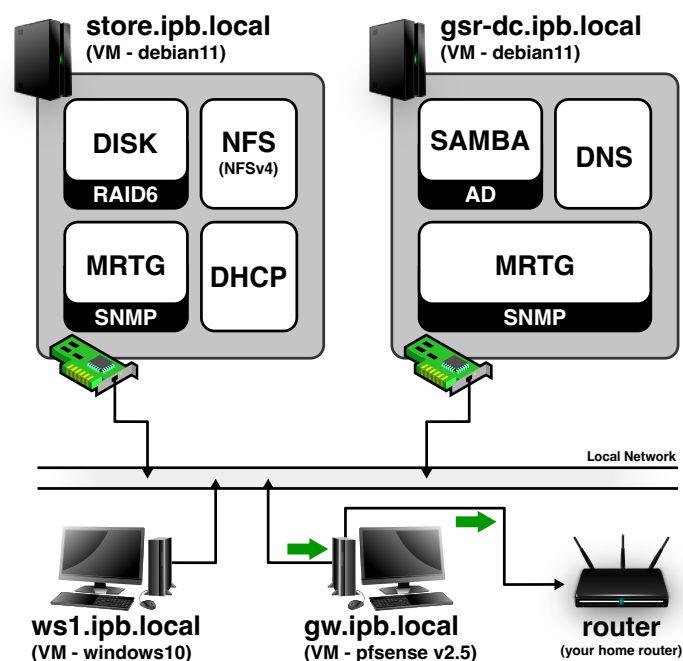
Level 5: Step 1
HARD
Theme: Final Integration
Due Date: 28/01/2023

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

File and Account Server

1. (*) Install and configure the necessary virtual machines and services in order to provide the configuration proposed below, having in mind the needs of a Small and Medium Enterprise (SME):



Hint: you can re-use any virtual machines from the previous submitted levels/steps

The overall architecture is constituted by three virtual machines:

(a) **gsr-dc.ipb.local** - VM with **Linux Debian 11** used for **User Account** and **Domain Server**. Providing:

- A **SAMBA** server with a working **Active Directory** for the **gsr.ipb.local** domain (**NETBIOS**: GSR)
- A correctly configured **DNS Server** with on Resource Record for each server in the network (forward and reverse).

Include the forward and reverse zone files in the report.

⚠ Don't use screenshots, use text!

- A shared folder `\\gsr-dc.ipb.local\gsr` accessible (read/write) by all domain users
- User accounts for the two members of the working group
- Disk Space and CPU should be monitored through **MRTG** using the data provided by **SNMP**.

(b) **store.ipb.local** - VM with **Linux Debian 11** used for storage. Providing:

- A **RAID6** array supported by set of **disks** to store files (any size).
- The disk should be accessible to **gsr-dc.ipb.local** via **NFS** (**NFSv4**)
- A properly configured **DHCP** service.
- A properly configured **Net-SNMP** service (with v3 authentication)
- Disk Space and CPU should be monitored through **MRTG** using the data provided by **SNMP**.

(c) **ws1.ipb.local** - VM with **Windows 10**:

- The workstation should join to the **gsr.ipb.local** domain
- Use the DNS server on **gsr.ipb.local** for name resolution
- The IP address should be offered by the **DHCP** server installed on **store.ipb.local**

(d) **gw.ipb.local** - VM with **pfsense v2.5**:

- Configure pfsense as a gateway for your local network
- Make the necessary configuration to provide VPN access to your network (**Open-VPN**)

2. Explain your local network topology, providing:

- (a) The possible subnet, netmask and gateway
- (b) The IP addresses (where applicable) for all hosts (and the method of configuration)

Enumerate any additional software, distributions and devices you would use to implement the requested features

Deliverables

3. All deliverables submitted via [discord#assignments-submission](#)):

1. PDF file with the report
2. The file name should be `15s1h_aXXXXX_aYYYYY.pdf`. This code means:

- Level 5
- Step 1
- Normal difficulty
- Students that participated in the group
- **Maximum:** 2 students